Raffles Institution
Physics Department

Practical WS18: Current in Parallel Wires & Discharging of Capacitor **Time: 1 hr 15 min**

Name: SOLUTIONS ( )

Class: 26S **Date:**

[2020/P4/Q1 modified]

1 In this experiment, you will investigate the current in an electrical circuit.

(a) You have been provided with two metre rules A and B, each with a resistance wire attached.

Take measurements to determine the resistance per unit length of each of the wires.

The resistance per unit length of the wire attached to rule A is R_A.

The resistance per unit length of the wire attached to rule A is R_B.

Resistance of 1.000 m of wire A and B are $18.1\,\Omega$ and $28.4\,\Omega$, respectively.

R_A = $18.1\,\Omega\,m^{-1}$

R_B = $28.4\,\Omega\,m^{-1}$ [2]

Use the digital multimeter as an ohmmeter to measure the resistance across 1.000 m of each wire.

(b) Set up the circuit as shown in Fig. 1.1.

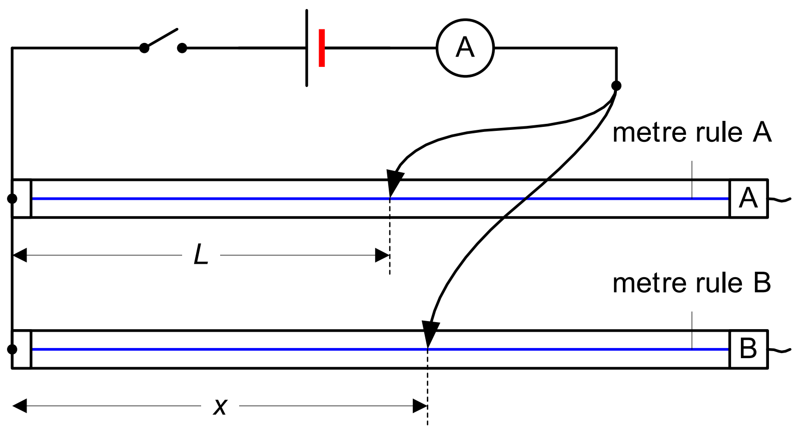


Fig. 1.1

L should be approximately half the length of the rule and *x* should be **greater** than *L*.

Close the switch.

Measure and record *L*, *x* and the ammeter reading *I*.

L = 0.500 m

x = 0.550 m

I = 0.230 A [2]

(c) Vary x, obtaining a suitable range of values between 0 cm and 100 cm, and repeat **(b)**, keeping L constant throughout.

x / m	I / A	$\frac{1}{x}$ / m^{-1}
0.550	0.230	1.82
0.600	0.224	1.67
0.650	0.221	1.54
0.750	0.212	1.33
0.800	0.209	1.25
0.900	0.203	1.11

Total 6 sets of readings

Range of x should be large enough but it must be larger than 0.500 m (following instructions from (b))

There is no need to take repeated measurements.

[3]

(d) It is suggested that I and x are related by the expression

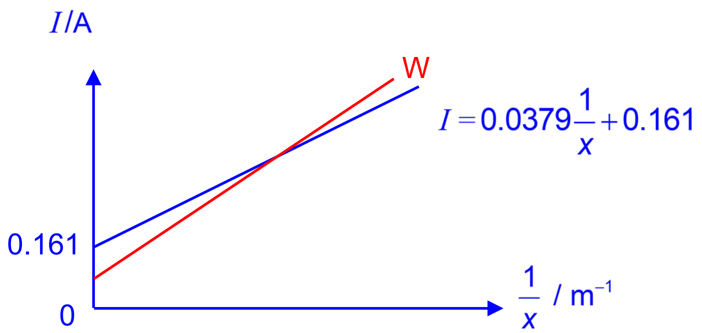
$$I = \frac{E}{R_A L} + \frac{E}{R_B x}$$

where E is the electromotive force (e.m.f.) of the cell.

Plot a suitable graph using a spreadsheet to determine a value for E.

Sketch the graph in the space below. Write the equation of the trendline.

Plot a graph of I against $\frac{1}{x}$. where the gradient is $\frac{E}{R_B}$ and the vertical intercept is $\frac{E}{R_A L}$.



Gradient $= \dfrac{E}{R_B}$

$E = \text{gradient} \times R_B = 0.0379 \times 28.4 = 1.08$ V

Alternatively:

Vertical intercept $= \dfrac{E}{R_A L}$

$E = \text{vertical intercept} \times R_A \times L = 0.161 \times 18.1 \times 0.500 = 1.48$ V

Actual value of E = 1.55 V

E = 1.08 V [3]

(e) **(i)** The experiment is now repeated with x measured on metre rule A and the same L in **(b)** measured on metre rule B.

Without taking further readings, sketch a line on your graph to show the results you would expect. Label this line W.

Give an explanation for W.

The gradient of the new line would now be $\dfrac{E}{R_A}$ while the vertical intercept is $\dfrac{E}{R_B L}$.

Since $R_A < R_B$, the gradient would be larger and vertical intercept smaller for line W. [2]

(ii) State and explain the value of $\dfrac{1}{x}$ when the 2 lines intersect.

The same current would flow through the circuit when $x = L = 0.500$ m

Hence, $\dfrac{1}{x} = 2.00\ \text{m}^{-1}$. [2]

[14 marks]

2 You will investigate the discharging of a capacitor.

(a) Set up the circuit as shown in Fig. 2.1, taking care to connect the capacitor the right way round.

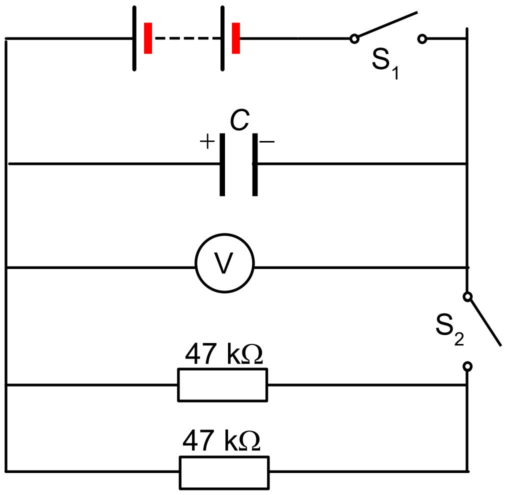


Fig 2.1

(i) Record the effective resistance R.

$$R = \frac{47}{2} = 23.5 = 24\ \text{k}\Omega$$

R = 24 kΩ .... [1]

(ii) Closed switch S_1 and wait for the capacitor to be fully charged.

Record the potential difference V_0 across the capacitor.

V_0 = 3.20 V .... [1]

(iii) Open switch S_1.

(iv) Close S_2 and immediately start the stopwatch.

Record V every 20 s from time $t = 0$ to $t = 200$ s. Tabulate your results.

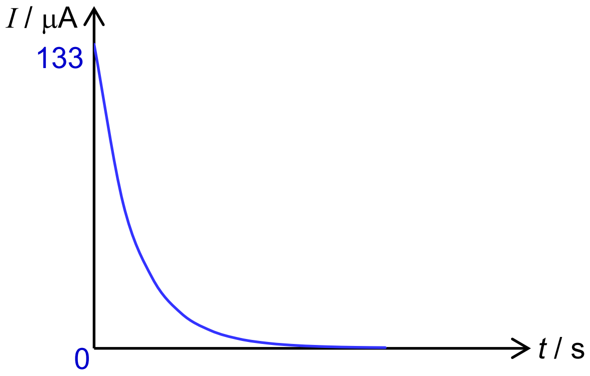
t / s	V_1 / V	V_2 / V	$\langle V \rangle$ / V	I / μA	Q_T / mC
0.0	3.20	3.20	3.20	133	0.00
20.0	1.522	1.520	1.521	63.4	1.97
40.0	0.747	0.732	0.740	30.8	2.91
60.0	0.369	0.364	0.367	15.3	3.37
80.0	0.190	0.186	0.188	7.83	3.60
100.0	0.100	0.096	0.098	4.08	3.72
120.0	0.056	0.050	0.053	2.21	3.78
140.0	0.031	0.028	0.030	1.23	3.82
160.0	0.019	0.016	0.018	0.729	3.84
180.0	0.012	0.010	0.011	0.458	3.85
200.0	0.009	0.007	0.008	0.33	3.9

Note:
- I and Q_T are calculated values.
- $I = \dfrac{\langle V \rangle}{R}$; Additional sf has been applied to the entire column for I.

[3]

(b) Plot a graph of discharge current I in the circuit with time t using a spreadsheet.

(i) Sketch the graph in the space below. [1]



(ii) Use your graph to estimate the time at which the current drops to 25% of its initial value.

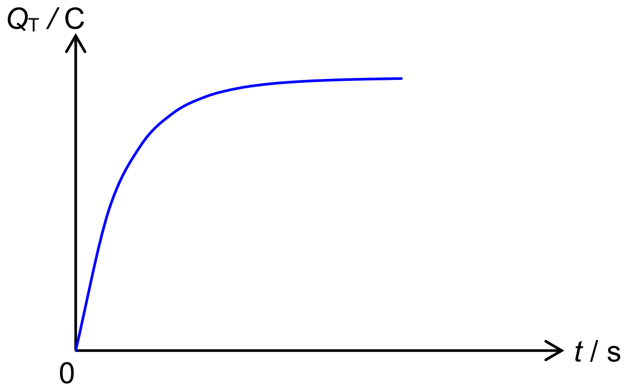
Adjust the limits of both vertical and horizontal axes to focus on the region around 33 mA.

Turn on major gridlines and select an appropriate scale to enable precise identification of 33.3 mA and the corresponding time t.

t = 37.9 s .... [1]

(c) Plot a graph of total charge Q_T that flowed through the circuit during discharge with time t using a spreadsheet.

(i) Sketch the graph in the space below. [1]



(ii) State the total charge Q_T that flowed through the circuit during discharge.

Value of Q_T when t = 200s.

Q_T = 3.9×10^{-3} C [1]

(iii) Measure and record the capacitance of C.

Calculate the initial charge Q_i.

$C = 1121\ \mu\text{F}$ (DMM can read to 0.1 uF but there's fluctuation)

$Q_i = CV_0 = (1121 \times 10^{-6})(3.20) = 3.6 \times 10^{-3}$ C

C = $1121\ \mu\text{F}$ [1]

Q_i = 3.59×10^{-3} C [1]

(iv) Comment on your values in **(c)(ii)** and **(c)(iii)**, and account for any difference.

The total charge Q_T is determined from the area under the I–t graph using trapezium rule. Due to shape of the graph in (b)(i), the trapezium rule produces an overestimate of the true area. Hence, Q_T is greater than the actual value Q_i.

[1]

[Total: 12]

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